

Hritik Kumar

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Professional Summary

AI-focused Information Technology undergraduate experienced in building AI applications using LLMs, AI agents, computer vision, and scalable backend systems. Skilled in FastAPI, React, vector databases, and multimodal AI workflows.

SKILLS

Languages: C/C++, Python, JavaScript, SQL

Frameworks: FastAPI, Flask, React, Tailwind, LangChain

AI/ML: Generative AI, LLMs, AI Agents, RAG, NLP, Computer Vision, OpenCV, HuggingFace, PyTorch, TensorFlow, Prompt Engineering, Multimodal AI

AI Workflows: Tool Calling, Function Calling, Retrieval Pipelines, Vector Search

Databases: MySQL, Firebase, Supabase, Pinecone, Qdrant

Tools: Git, GitHub, VS Code, ZeroMQ, ONNX

Projects

NavDrishti — AI-Powered Smart Cane for Visually Impaired — Python, Raspberry Pi, YOLOv26l, FastAPI, WebSockets, ZeroMQ, TTS, React [GitHub]

- o Built real-time edge AI pipeline (YOLO detection → contextual processing → TTS feedback) on Raspberry Pi 4, achieving sub-second latency for continuous environmental perception across diverse real-world scenarios.
- o Designed distributed multi-worker inference system (ZeroMQ + FastAPI + WebSockets) and trained multiple specialized CV models (general navigation, road-crossing, currency detection, pothole detection), enabling parallel vision tasks and context-aware perception under hardware constraints.

DualCast — Real-Time AI eSports Commentary Engine — Python, FastAPI, WebRTC, Qwen2.5-VL, YOLO, ResNet, ElevenLabs [GitHub]

- o Engineered low-latency multimodal pipeline (Qwen2.5-VL + YOLO + ResNet) processing live gameplay streams via WebRTC, achieving real-time event understanding at 30 FPS.
- o Built dual-agent LLM system (Hype + Analyst) with synchronized voice synthesis using ElevenLabs, powered by asynchronous multi-model inference for continuous real-time commentary.

Kala Setu — AI-powered Artisan Assistance Platform — FastAPI, React, Gemini AI, NLP, Supabase, Pinecone, Qdrant

- o Built AI-powered e-commerce platform enabling artisans to create listings, price products, and market crafts using voice input and LLM-based content generation (titles, descriptions, SEO tags).
- o Developed AI chatbot workflows using Gemini API, Pinecone, and Qdrant for semantic retrieval, contextual response generation, and vector-based search.

Experience

Freelance Full Stack Developer 2025

- o Built and deployed production-ready web applications using React + FastAPI with authentication and database integration.
- o Integrated backend APIs and AI-driven features to automate workflows and enhance user experience.

Leadership

- o **Treasurer, Alumni Committee** — Managed budgeting, financial planning, and fund allocation for institutional events and activities.
- o **Tech Lead, IEEE** — Led technical initiatives, mentored team members, and organized development-focused projects and events.

Achievements

1st Place — Vihaan 9.0 Hackathon, DTU 2026

1st Place (Theme : Sustainability) — Tech Aarambh Hackathon, ACEM Faridabad 2026

1st Place — LOCKED-IN Hackathon, GGSIPU USICT 2026

Education

B.Tech in Information Technology CGPA: 8.01

Guru Gobind Singh Indraprastha University, USICT